

What is Internet?

The interconnected network of computer networks that spans the globe.

What are the reasons for Internet Growth in the 1990s

- Removal of the ban on commercial activity
- Development of the World Wide Web by Tim Berners-Lee at CERN
- Development of Mosaic, the first graphics-based web browser at NCSA
- Personal computers were increasingly available and affordable
- Online service providers offered low-cost connections to the Internet

Who runs the Internet?

You may be surprised that there is no single person “in charge” of the global interconnected network of computer networks known as the Internet. Instead, Internet infrastructure standards are overseen by groups, such as the Internet Engineering Task Force (IETF) and the Internet Architecture Board (IAB).

What is the World Wide Web?

The graphical user interface to information stored on computers running web servers connected to the Internet.

What are the Internet Standards & Coordination

IETF – Internet Engineering Task Force

The principal body engaged in the development of new Internet protocol standard specifications.

RFC – Requests for Comments

A formal document from the IETF that is drafted by a committee and subsequently reviewed by interested parties

IAB – Internet Architecture Board

Provides guidance and broad direction to the IETF. Responsible for publications for RFCs.

- ICANN - The Internet Corporation for Assigned Numbers & Names
 - Non-profit organization

- Main function is to coordinate the assignment of:
 - Internet domain names
 - IP address numbers
 - Protocol parameters
 - Protocol port numbers.

Growth of the Internet

Year	Percentage of Global Population Using the Internet
1995	0.4%
2000	5.8%
2005	15.7%
2010	28.10%
2015	45%
2018	55.1%
2019	56.1%

Source: <http://www.internetworldstats.com/emarketing.htm>

Differentiate between Intranet & Extranets

Intranet

- A private network contained within an organization or business used to share information and resources among coworkers.

Extranet

- A private network that securely shares part of an organization's information or operations with external partners

Web Standards and the W3C Consortium

- W3C – World Wide Web Consortium
 - Develops recommendations and prototype technologies related to the Web
 - Produces specifications, called Recommendations, in an effort to standardize web technologies
 - WAI – Web Accessibility Initiative

Reliability & Information on the Web

Questions to Ask:

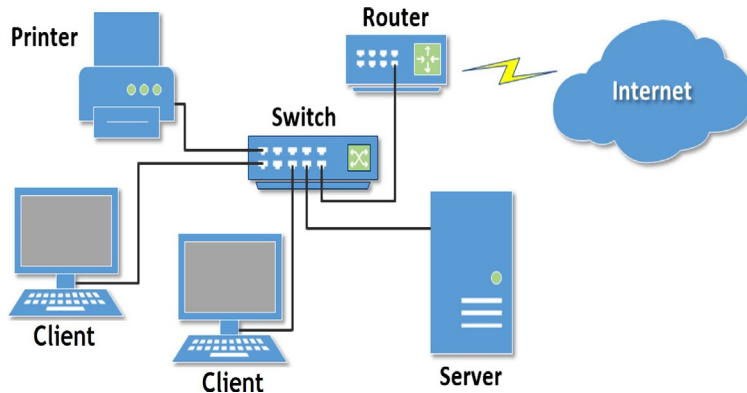
- Is the organization credible?
- How recent is the information?
- Are there links to additional resources?
- Is it Wikipedia? *If so, further research is needed.*

Network Overview

What is a Network

two or more computers connected together for the purpose of communicating and sharing resources

Figure 1.4 Common components of a network



Types of Networks

LAN – Local Area Network

- Usually confined to a single building or group of buildings

WAN – Wide Area Network

- Usually uses some form of public or commercial communications network to connect computers is widely dispersed geographical areas.

Internet Infrastructure

What is Internet Backbone?

A high-capacity communication link that carries data gathered from smaller links that interconnect with it.

The Client/Server Model

Client/Server can describe a relationship between two computer programs—the “**client**” and the “**server**”.

Client: requests for some type of service (such as a file or database access) from the server.

Server: fulfills the request and transmits the results to the client over a network

Client – Web Browser Server – Web Server



Figure 1.5 Web client and web server

Web Client

Connected to the Internet when needed

Usually runs web browser (client) software (*such as Microsoft Edge or Google Chrome*)

Uses HTTP (Hypertext Transfer Protocol) or HTTPS Requests web pages from server

Receives web pages and files from server

Web Server

Continually connected to the Internet Runs web server software (*such as Apache or Internet Information Server*) Uses HTTP (Hypertext Transfer Protocol) or HTTPS

Receives request for the web page

Responds to request and transmits status code, web page, and associated files

MIME Type

Multi-Purpose Internet Mail Extension

- a set of rules that allow multimedia documents to be exchanged among many different computer systems

Internet Protocols

Protocols

- Rules that describe the methods used for clients and servers to communicate with each other over a network.
- There is no single protocol that makes the Internet and Web work.
- A number of protocols with specific functions are needed.

FTP - File Transfer Protocol

A set of rules that allow files to be exchanged between computers on the Internet. Web developers commonly use FTP to transfer web page files from their computers to web servers. FTP is also used to download programs and files from other servers to individual computers.

E-mail Protocols

Sending E-mail

- SMTP Simple Mail Transfer Protocol

Receiving E-mail

- POP (POP3) Post Office Protocol
- IMAP Internet Mail Access Protocol

HTTP - Hypertext Transfer Protocol

- A set of rules for exchanging files such as text, graphic images, sound, video, and other multimedia files on the Web.
- Web browsers send HTTP requests for web pages and their associated files.
- Web servers send HTTP responses and the requested files back to the web browsers.

HTTPS – Hypertext Transfer Protocol Secure: Combines HTTP with a security and encryption protocol

TCP/IP Transmission Control Protocol/ Internet Protocol

TCP/IP has been adopted as the official communication protocol of the Internet.

TCP and IP have different functions that work together to ensure reliable communication over the Internet.

TCP Transmission Control Protocol

Purpose is to ensure the integrity of communication

Breaks files and messages into individual units called packets

Figure 1.6 TCP packet



IP Internet Protocol

- A set of rules that controls how data is sent between computers on the Internet.
- IP routes a packet to the correct destination address.
- The packet gets successively forwarded to the next closest router (a hardware device designed to move network traffic) until it reaches its destination.
<http://www.monitis.com/traceroute/>

What is an IP Address?

Each device connected to the Internet has a unique numeric IP address. These addresses consist of a set of four groups of numbers, called octets.

216.58.194.46 will get you Google!

An IP address may correspond to a domain name.

Domain Name

- Locates an organization or other entity on the Internet
- Domain Name System
 - Divides the Internet into logical groups and understandable names
 - Associates unique computer IP Addresses with the text-based domain names you type into a web browser
 - Browser: <http://google.com>
 - IP Address: 216.58.194.46

Uniform Resource Identifier

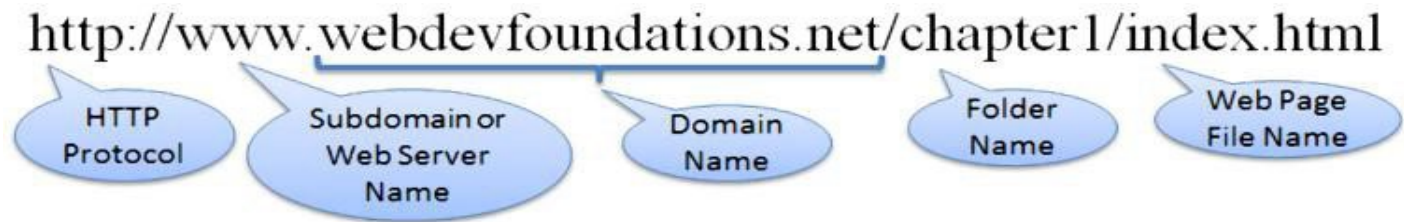
URI – Uniform Resource Identifier

- identifies a resource on the Internet

URL – Uniform Resource Locator

- a type of URI which represents the network location of a resource such as a web page, a graphic file, or an MP3 file.

Figure 1.8 URL Describing a file within a folder



TLD Top-Level Domain Name

A top-level domain (TLD) identifies the right-most part of the domain name.

Examples of generic TLDs:

.com, .org, .net, .mil, .gov, .edu, .int, .aero, .asia, .cat,
.jobs, .name, .biz, .museum,
.info, .coop, .post, .pro, .tel, .travel

Country Code TLDs

Two character codes originally intended to indicate the geographical location (country) of the web site.

In practice, it is fairly easy to obtain a domain name with a country code TLD that is not local to the registrant.

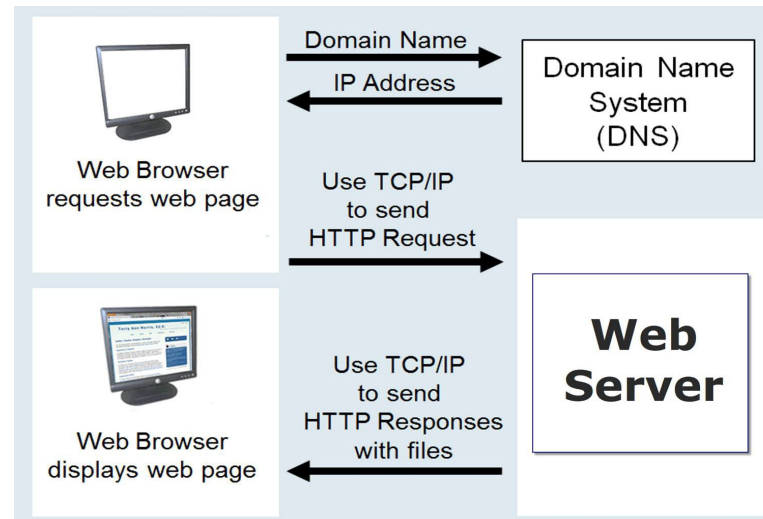
Examples:

- .tv, .ws, .au, .jp, .uk
- See <http://www.iana.org/cctld/cctld-whois.htm>

Domain Name System

The Domain Name System (DNS) associates Domain Names with IP addresses.

Figure 1.9 Accessing a web page



Markup Languages

SGML – Standard Generalized Markup Language

- A standard for specifying a markup language or tag set

HTML – Hypertext Markup Language

- The set of markup symbols or codes placed in a file intended for display on a web browser.

Markup Languages (2)

XML – eXtensible Markup Language

- A text-based language designed to describe, deliver, and exchange structured information.
- It is not intended to replace HTML – it is intended to extend the power of HTML by separating data from presentation.

Markup Languages (3)

XHTML—eXtensible Hypertext Markup Language

- Developed by the W3C as the reformulation of HTML 4.0 as an application of XML.
- It combines the formatting strengths of HTML 4.0 and the data structure and extensibility strengths of XML.

Markup Languages (4)

HTML 5

- The next version of HTML4 and XHTML
- <https://www.w3.org/TR/html5/>
- It's already been updated!
- HTML 5.1

— <https://www.w3.org/TR/html51/>

- HTML 5.2

— <https://www.w3.org/TR/html52/>

Popular Uses of the Internet

E-Commerce Mobile Access Blogs

Wikis

Social Networking RSS

Podcasts

Cloud Computing